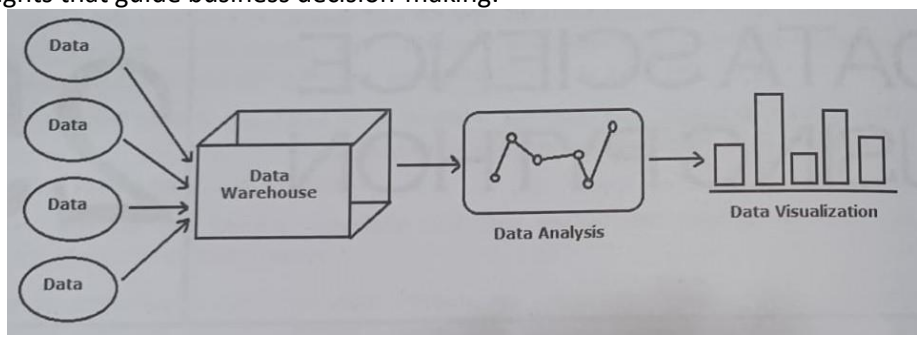
Unit 5- Overview of Data Science

Data science definition

Data science is the study of data. Data scientists construct questions around specific data sets and then use data analytics and advanced analytics to find patterns, create predictive models, and develop insights that guide business decision-making.



Introduction

Data science is the study of data to extract meaningful insights for business. Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization's data. These insights can be used to guide decision making and strategic planning.

This analysis helps data scientists to ask and answer questions like what happened, why it happened, what will happen, and what can be done with the results.

The accelerating volume of data sources, and subsequently data, has made data science is one of the fastest growing field across every industry.

Role of Data Science in various industries

Data science is being applied to several industries with transformative results. Operational efficiencies are significantly increasing, leading to increases in revenue margins.

1). Health Care and Pharmaceuticals

The health care industry includes a massive amount of data from the Electronic Health Records (EHR), genome sequencing, mobile health devices, social media, and other sources.

This information, with the application of data science, is leading to important improvements in clinical care and treatments.

Computer vision, an application of artificial intelligence to train computers to see images like a human, acts as a second eye to improve the accuracy of clinical decisions.

In health care it leverages data from smartphones, wearable sensor devices, and other patient monitoring tools for improvements in personalized patient care and detecting life-threatening disease.

2). Finance

Applications of data science to finance are important to financial security, managing risk, marketing and improving trading. The communications that banks rapidly send after a suspicious transaction are made possible through machine learning.

Illegal insider trading can be caught faster by using deep learning to analyze trading patterns before and after the announcement of significant company news.

Banks avoid human error by applying predictive analytics to loan history, bank transaction, income, demographics and even data from social media to guide loan assessments and to detect early warning signs in existing loan service

3). Retail and eCommerce

Suggestions for new products based on a customer's shopping behavior are one of the most recognizable impacts of data science in retail and e-commerce.

Similarly market basket analysis, also driven by machine learning and deep learning, informs future purchases based on customer data.

Data science helps companies set optimal prices through algorithms that analyze pricing flexibility, customer and geographic data, and other criteria.

Machine learning also helps companies manage inventory by analyzing purchase patterns to develop strategies too.

4). Manufacturing

Delivering products on time and in the right quantities is essential to manufacturing.

Data science plays an important role for the industry by improving efficiency and reducing costs.

Data Science can reduce unplanned downtime, due to machine maintenance.

Companies are also using computer vision to improve on human quality control to avoid using defective parts.

Sales forecasting is important to manufacturing companies and requires predicting manufacturing volumes.

5). Government

Analyzing spending data to identify and prevent waste, fraud, and abuse

Using business intelligence to help government at all levels make better financial decisions

Working with researchers in studies and tracking findings that could determine the course of new drugs and consumer goods

Collecting and analyzing defense intelligence to improve defense systems

Working with state and local departments to coordinate data gathering, analysis, and modeling

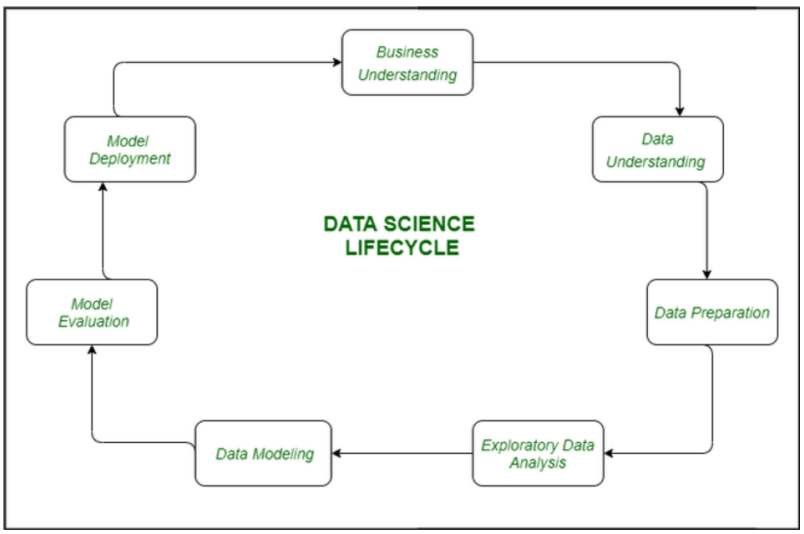
Data Science Process and Life Cycle

-You can have data without information, but you cannot have information without data.

-Data can be proved to be very much useful if we know how to manipulate it.

-This logic behind the data or the process behind the manipulation is what is known as Data Science.

-Some steps are necessary for any of the tasks that are being done in the field of data science to derive any meaningful results from the data.



1). Data Collection:
After formulating any problem statement, the main task is to calculate data that can help us in our analysis and manipulation. Sometimes data is collected by performing survey.
2). Data Cleaning:
Most of the real-world data is not structured and requires cleaning and conversion into structured data before it can be used for any analysis or modeling.
3). Exploratory Data Analysis:
This is the step in which we try to find the hidden patterns in the data. Also, we try to analyze different factors which affect the target variable and the extent to which it does so. How the independent features are related to each other and what can be done to achieve the desired results all these answers can be extracted from this process as well. This also gives us a direction in which we should work to get started with the modeling process.
4). Model Building:
Different types of machine learning algorithms as well as techniques have been developed which can easily identify complex patterns in the data which will be a very tedious task to be done by a human.
5). Model Deployment:
After a model is developed and gives better results on the holdout or the real-world dataset then we deploy it and monitor its performance. This is the main part where we use our learning from the data to be applied in real-world applications and use cases.

Data Science tools and libraries
Data frame is an object that is useful in representing data in the form of rows and columns.
Sometimes data come in the form of Excel spreadsheet, list, tuple or even in the form of python dictionary.
We can represent that data in the form of a data frame. Once the data is stored into the data frame, we can perform various operations that are useful in analyzing and understanding the data.
We must download a package known as 'Pandas' to work with the data.
If we need to work with excel file, then we must download 'xlrd' package.

Working with .csv file using Pandas
Example: Read the csv file and display the data.

```
import pandas as pd
df=pd.read_csv("product.csv")
print(df)
```

Working with .xlsx file using Pandas

```
#Creating data frame from .xlsx files
        import pandas as pd
        import xlrd
        df=pd.read_excel("prod.xlsx")
        print(df)
```

Working with Python Dictionary using Pandas

```
#Creating data frame from Python Dictionary
        stud={'roll':[1,2,3,4,5],'name':['abc','cde','efg','ghi','ijk'],'city':['rajkot','surat','goa','diu','vadod
        ara']}
        import pandas as pd
        df=pd.DataFrame(stud)
        print(df)
```

Working with Python Tuples using Pandas

```
#Creating data frame from Tuples
        stud=[(1,'abc','rajkot'), (2,'cde','surat'), (3,'efg','goa'), (4,'ghi','diu'), (5,'ijk','vadodara')]
        import pandas as pd
        df=pd.DataFrame(stud, columns=['roll','name','city'])
        print(df)

#Creating data frame from Tuples
        stud=[(1,'abc','rajkot'), (2,'cde','surat'), (3,'efg','goa'), (4,'ghi','diu'), (5,'ijk','vadodara')]
        import pandas as pd
        df=pd.DataFrame(stud, columns=['roll','name','city'])
        print(df)
# Knowing number of rows and columns
        print(df.shape)
#Display first five records
        print(df.head())
#Display last five records
        print(df.tail())
#Display first two records
        print(df.head(2))
#Display last two records
        print(df.tail(2))
#Retrieving range of rows
        print(df[2:5])
#Retrieving data from columns
        print(df[['name','city']])
#Finding Maximum values
        print(df['roll'].max())
#Finding Minimum values
        print(df['roll'].min())
#Display students who belongs to rajkot
        print(df[df.city=='rajkot'])
#Display students whose roll is bigger than 2.
        print(df[df.roll>2])

#Display students' name who belongs to rajkot
        print(df[['name']][df.city=='rajkot'])
```

```
#Handling missing data
import pandas as pd
import xlrd
df=pd.read_excel("prod.xlsx")
print(df)
new=df.fillna({'prod_nm':'Prod name missing','Category':'Category is missing',2023:'Sales of unit is missing'})
print(new)
```

Data Visualization

-When the analyzed data is shown in the form of text (whether it may be in the form of no. of statements or paragraphs) it consumes so much of time to make the document and even it is more time consuming to read the document and understand it (without bias).

-The better option is to use diagrams such as bar graph, pie chart, line chart or any other charts available, which makes better understanding (without bias) and less time consuming.

-The data represented in the form of pictures or charts/graphs is known as data visualization.

-To create such chars/graphs in python we use 'matplotlib' module.

Bar graph

-As per the name implies, the bar graph represents data in the form of vertical or horizontal bars.

-The bar graph is used to compare the quantities.

```
#Program to display bar chart for the different types of books in the book stall.
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_excel('bookstall.xlsx')
print(df)
#Extracting the type_of_book and sales into x and y
x=df['Type_of_book']
y=df['Sales']
#Creating the bar graph using bar() function
plt.bar(x,y, label='Category_wise_sales_of_Books',color='Blue')
#Setting the label for x-axis and y-axis
plt.xlabel('Type_of_Book')
plt.ylabel('Sales')
#Setting the title of the ownership
plt.title('Atmiya Book Shop')
#Showing the legend
plt.legend()
#To display the graph
plt.show()
```

#Pie Chart
#To draw a pie chart representing sales in percentage of four different products of the company.

```
import matplotlib.pyplot as plt
#Taking the percentage of sales of different products
sales_per=[35,10,25,30]
#Taking the names of the products
prod_name=['Hatch back','Sedan','Premium','Commercial']
#Giving the color to each product category
col=['red','pink','blue','yellow']
#Creating the pie chart (Using pie())
plt.pie(sales_per,labels=prod_name,colors=col,startangle=90)
#Setting upt the titles and legend
plt.title('TATA Motors')
plt.legend()
#Showing the chart
plt.show()
```

#Line graph
#To draw a line chart to represent population increase in last five years

```
import matplotlib.pyplot as plt
years=['2023','2022','2021','2020','2019']
incr_pop=[70,45,60,75,30]
#Creating the line graph using plot()
plt.plot(years,incr_pop,color='red')
#Setting title and labels
plt.title('Population growth of India')
plt.xlabel('Years')
plt.ylabel('Increase_of_Population_in Lakhs')
#Display the line chart
plt.show()
```

```
#Using Numpy
#Grouped Bar graph
        import pandas as pd
        import matplotlib.pyplot as plt
        import numpy as np
        df=pd.read_excel('stationary.xlsx')
        print(df)
        #Extracting the type_of_book and sales into x and y
        year=[2021,2022,2023]
        x=df['Pencil']
        x1=df['Pen']
        # The np.arange( ) function from numpy library is used to create a range of values.
        #We are creating the X-axis values depending on the number of groups in our example.
        xaxis=np.arange(len(year))
        #Creating the bar graph using bar() function (even we can use barh() to display bar
        horizontally)
        # To avoid overlapping of bars in each group, the bars are shifted -0.2 units and +0.2 units
        from the X-axis.
        # The width of the bars of each group is taken as 0.4 units.
        plt.bar(xaxis - 0.2,x,0.4, label='Pencil')
        plt.bar(xaxis + 0.2,x1,0.4, label='Pen')
        # Set the current tick location
        plt.xticks(xaxis, year)
        #Setting the label for x-axis and y-axis
        plt.xlabel('Product')
        plt.ylabel('Sales')
        #Setting the title of the ownership
        plt.title('Atmiya Stationary')
        #Showing the legend
        plt.legend()
        #To display the graph
        plt.show()


#Draw a Scatter Plot for random numbers
        import matplotlib.pyplot as plt
        import numpy as np

        #x and y must be same size
        #Random is a sub-module of module numpy
        #Randn() generates random decimal values
        x = np.random.randn(10)
        print(x)
        y = np.random.randn(10)
        print(x)
        plt.scatter(x,y)
        plt.xlabel('x-axis')
        plt.ylabel('y-axis')
        plt.title("Scatter diagram ")
        plt.show()
```

```
#Create an area plot
import numpy as np
import matplotlib.pyplot as plt
#create data
x = range(1,6)
y = [1,4,6,2,4]
#area plot
plt.fill_between(x,y)
plt.title('Area plot')
plt.show()
```

Career opportunities in Data Science

-According to International Data Corporation (IDC), by 2025, global data will grow to 175 zettabytes.
-Data Science enables companies to efficiently understand gigantic data from multiple sources and derive valuable insights to make smarter data-driven decisions.
-Data Science is widely used in various industry domains, including marketing, healthcare, finance, Production, Government, and many more.
-So, lots of career opportunities lies in the field of Data Science.
-One can opt for any of the roles discussed below as per his/her choice and eligibility.

1). Data Scientist:

-Data scientists must be a mathematician, a computer scientist, and a business strategist all at once, necessitating expertise in several different disciplines.
-This complex skill set means that data scientists need to consistently have one foot in the information technology sector, and another planted firmly in the business world.
-That's part of what makes them in such high demand.

2). Data Analyst:

-Data analysts are most often responsible for transforming data sets into usable forms, such as reports or presentations.
-Depending on the industry, this may involve gleaning insight from consumer data sets, making strategic recommendations based on dense financial data, or simply organizing messy data into a more accessible format.
-A qualified data analyst will have a solid understanding of R, Python, HTML, C/C++ and SQL.

3). Data Engineer:

-Data engineers are the designers, builders, and managers of the information or big data infrastructure.
-They assist in developing the architecture that helps analyze and process data in a manner best suited for their organization.
-It is their role to also make sure those systems are performing smoothly.
-A data engineer has a background in software engineering as well as skills in the following languages: SQL, HIVe, Pig, R, Matlab, SAS, SPSS, Python, Java, and Ruby.

4). Data Analyst:

-The role of the business analyst is to improve business processes by serving as a liaison between business and IT with a clear directive to focus on advancing strategic business objectives.
-Most business analysts are focused on producing usable deliverables, such as reports and presentations, that can be easily understood by others in the organization who are not data scientists themselves.
-Business analysts possess the basic skills of data visualization tools and data modeling, however their educational background is in business.

4). Data Architect:

-Data architect roles are becoming more common.
-This position creates the blueprints for data management systems to integrate, centralize, protect, and maintain data sources.

-They understand the languages of SQL, XML, Hive, Pig, and Spark as well as the skills of warehouse solutions, systems development, and database architecture.

Prerequisites and recommended skills for Data Science

1. Understanding of Data
2. Understanding of Algorithms/Logic
3. Understanding of Programming
4. Understanding of Statistics
5. Understanding of business domain

Recommended skills for Data Science

1. Statistics
2. Maths
3. Designing and Understanding the logic
4. Programming languages
5. Data bases.